

Nikhil Ankush Thange

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Professional Summary

Full Stack Software Engineer specializing in high-concurrency microservices, reactive React 19/TypeScript architectures, and computer vision AI pipelines. Engineered distributed backends sustaining **500+ concurrent requests (sub-50ms P95)** with Redis/MongoDB and deployed production applications on **AWS**. Dedicated to writing clean, test-driven code (**Jest/Vitest**) and automating CI/CD release cycles.

Technical Skills

Languages & Core: TypeScript, JavaScript (ES6+), Java, SQL, Python, HTML5, CSS3, Data Structures & Algorithms, OOP

Frontend Stack: React 19, Next.js, Redux Toolkit, TanStack Query, Tailwind CSS, WebSockets, Canvas API, PWA, Vitest

Backend & Microservices: Node.js, Express.js, RESTful APIs, OpenAPI/Swagger, Socket.IO, PM2, Middleware, JWT, RBAC

Databases & Caching: MongoDB (2dsphere Geospatial, Aggregations), PostgreSQL, Redis (TTL Caching), Mongoose

Cloud, DevOps & Testing: AWS (EC2, S3, CloudFront), Docker, Nginx, GitHub Actions (CI/CD), Jest, Supertest, Postman, Git

Work Experience

Software Development Engineer Intern

Jul 2026 – Present

Chitralai | React 19, TypeScript, Node.js, Express, PostgreSQL, AWS, Jest, REST APIs

Remote

- **High-Converting Cloud Architecture:** Boosted user lead conversion rates by **25%** and slashed page load time by **40%** by architecting a responsive React 19 SPA with TanStack Query caching, deployed via **AWS (S3, CloudFront CDN, EC2)**.
- **Core Web Vitals Engineering:** Reduced Cumulative Layout Shift (**CLS**) by **30%** and cut First Contentful Paint (**FCP**) to **<800ms** by engineering the *ReelIt* video streaming feed with DOM virtualization and asynchronous lazy loading in Tailwind CSS.
- **High-Throughput Microservices:** Stabilized P95 REST API response latency to **sub-65ms** across 20+ endpoints by building Node.js/Express services with Redis caching, optimized **PostgreSQL** queries, and documented **OpenAPI/Swagger** schemas.
- **Automated Testing & CI/CD:** Decreased production defect escapes by **35%** by implementing automated unit/integration test suites using **Jest & Supertest (85%+ code coverage)**, integrated into **GitHub Actions CI/CD** pipelines in Agile sprints.

Featured Projects

BMC Smart Civic Operating System (CityOS)

[Live Demo](#) | [GitHub](#)

React 19, TypeScript, Node.js, Express, YOLOv8, PyTorch, Redis, MongoDB, Docker, Nginx, Leaflet GIS

- **Custom AI Vision Pipeline:** Fine-tuned a **YOLOv8 Vision Model** (PyTorch, CUDA) on 6 civic defect classes, attaining **91.4% mAP₅₀**; built an edge triage pipeline with client-side Laplacian blur rejection and active feedback re-training.
- **High-Concurrency Backend:** Scaled Node.js PM2 cluster instances with Redis TTL caching and Mongoose pooling (20–100), sustaining **500+ concurrent operations** at **sub-50ms P95** latency.
- **Geospatial Engine & Deduplication:** Implemented compound MongoDB **2dsphere** indexing and Turf.js Point-in-Polygon (PIP) algorithms across 24 Mumbai administrative wards, automating 35m spatial grievance deduplication.
- **PWA & Production DevOps:** Engineered a React 19 PWA with Service Worker offline sync, client-side Canvas compressor (**95% byte reduction**), multi-stage Docker containerization, and Nginx reverse proxy.

AI-Powered Interview & Recruitment Intelligence Platform

[Live Demo](#) | [GitHub](#)

React 19, Node.js, Express, NVIDIA NIM LLMs, Socket.IO, PostgreSQL, MongoDB, JWT, Tailwind CSS

- **Low-Latency LLM Streaming:** Architected real-time AI mock interview evaluations by streaming NVIDIA NIM LLM token responses over WebSockets (Socket.IO), reducing perceived evaluation latency to **<300ms**.
- **ATS Parser & Scoring Engine:** Engineered an automated resume parsing and candidate matching engine with structured prompt validation, improving candidate-job semantic alignment accuracy to **88%**.
- **Enterprise Security & RBAC:** Built multi-tenant role-based access control (Candidate, Recruiter, Admin) with stateless JWT authentication, cryptographic password hashing, and granular route middleware.

Honors & Technical Achievements

- **Quasar 4.0 Hackathon Finalist:** Selected among the top finalist teams for designing an AI-driven Civic-Tech distributed system.
- **IBM Full Stack Software Architecture:** Certified in Enterprise Microservices, RESTful API Design, and Cloud Deployment.
- **Open Source & Code Quality:** Maintained 100% test coverage and sub-100ms API benchmarks across open-source full-stack repositories.

Education

Bachelor of Engineering in Information Technology

CGPA: 7.50 / 10.0 | Aug 2023 – Jun 2027 (Expected)

Vasantdada Patil Pratishthan's College of Engineering (Mumbai University)

Mumbai, Maharashtra

Higher Secondary Certificate (HSC) – Science

61.0% | Jun 2021 – May 2023

Ramniranjan Jhunjhunwala College of Arts, Science & Commerce

Mumbai, Maharashtra

Secondary School Certificate (SSC)

87.0% | Jun 2020 – Mar 2021

Saraswati Vidya Niketan

Mumbai, Maharashtra